Lecture 30.12: Stacking Architecture & Out-Of-Fold (OOF) Logic











📌 Topic Overview

Comprehensive conceptual breakdown of Stacking (Stacked Generalization). Explores why simple voting is insufficient, how Level-1 Meta-Learners train on base model predictions, and why K-Fold Out-Of-Fold (OOF) splitting is strictly mandatory to prevent catastrophic data leakage.

📋 Key Learning Objectives

- Concept of Stacking (Stacked Generalization) by David Wolpert.
- Level-0 Base Learners vs Level-1 Meta-Learner architecture.
- The Data Leakage Catastrophe: Why base models cannot predict on their own training data.
- K-Fold Out-Of-Fold (OOF) Prediction Matrix generation step-by-step.
- Training the Meta-Model on the unbiased OOF feature matrix.
- Test set inference workflow: Retraining base models and predicting meta-features.

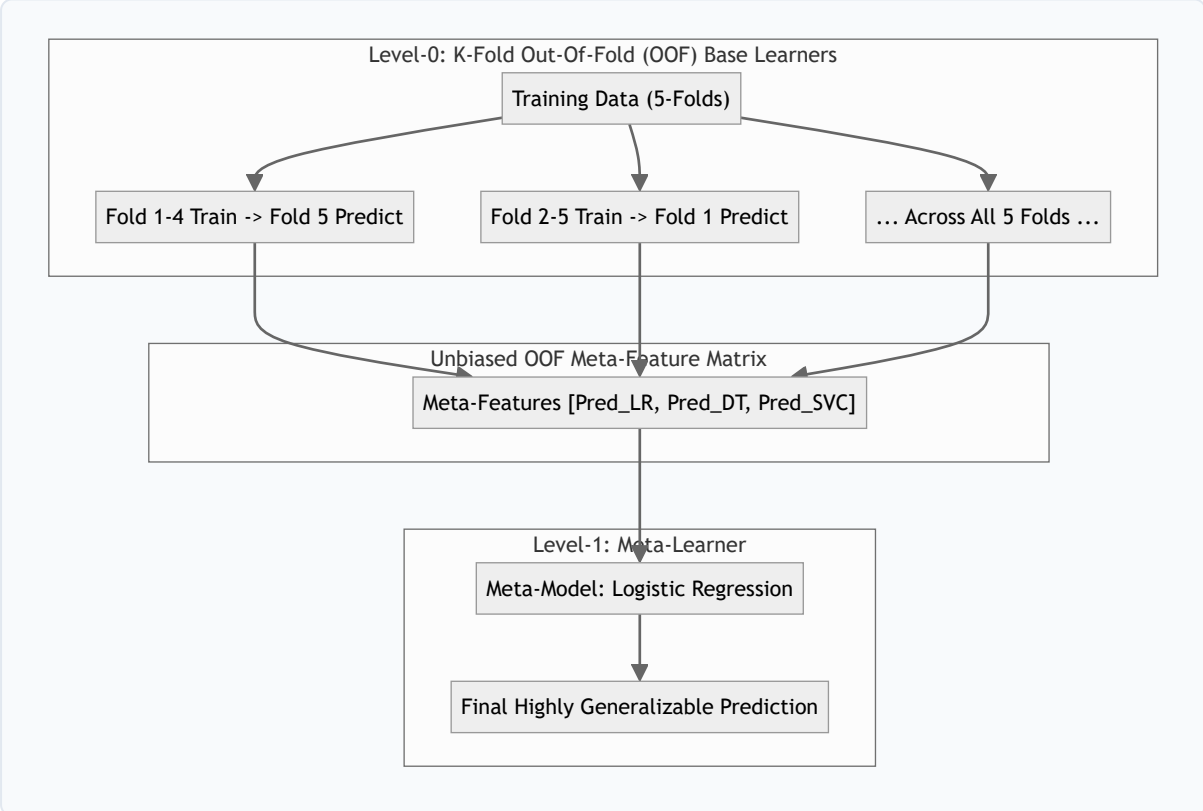
🔍 In-Depth Conceptual Breakdown

- **Stacking Ka Core Idea**: Voting mein hum sabhi models ko barabar ya fixed weight dete hain. Lekin **Stacking** mein hum ek naya model (**Meta-Model**) bitha dete hain jo yeh seekhta hai ki kaun se base model ki baat kab sunni hai aur kaise combine karna hai!

- **Level-0 vs Level-1**: Level-0 par multiple diverse models hote hain (**Logistic Regression**, **Decision Tree**, **SVM**). Level-1 par ek single **Meta-Model** hota hai (default: Logistic Regression ya Linear Regression).

- **Data Leakage Ka Bada Khatra**: Agar hum base models ko poore training data par train karein aur usi data par unse predictions lein, toh decision tree 100% accuracy de dega! Meta-model sochega ki Decision Tree bhagwan hai aur baaki models ko zero weight de dega. Test set par aate hi model fail ho jayega!
- **K-Fold Out-Of-Fold (OOF) Solution**:
- 1. Training data ko K folds (jaise 5 folds) mein baanto.
- 2. Har fold k ke liye, base models ko baaki 4 folds par train karo aur holdout fold k par predict karo.
- 3. In holdout predictions ko jod kar ek unbiased $N \times M$ meta-features matrix banayi jati hai jisme 0% data leakage hota hai!
- **Meta-Model Training**: Meta-model is OOF prediction matrix aur original target y par train hota hai.
- **Test Set Inference**: Test set ke liye saare base models poore training set par retrain hote hain, test set par predict karte hain, aur wo predictions meta-model ko di jati hain.

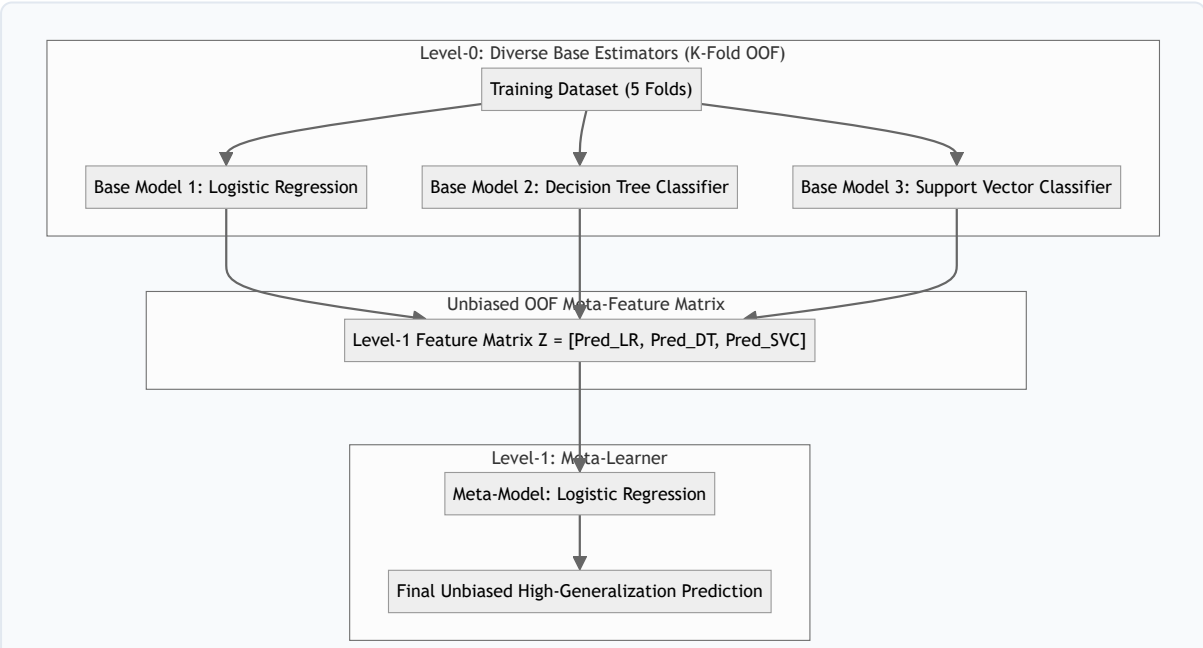
📊 Architectural Flowchart



📋 Ma'am's K-Fold Out-Of-Fold (OOF) Stacking Matrix

Fold ID	Training Folds for Base Models	Holdout Prediction Fold	Meta-Feature 1 ($Pred_{LR}$)	Meta-Feature 2 ($Pred_{DT}$)	Meta-Feature 3 ($Pred_{SVC}$)	Target Label (y)
Fold 1	Folds 2, 3, 4, 5	**Fold 1 Rows**	$\hat{y}_{1,LR}$	$\hat{y}_{1,DT}$	$\hat{y}_{1,SVC}$	$y_{\text{Fold 1}}$
Fold 2	Folds 1, 3, 4, 5	**Fold 2 Rows**	$\hat{y}_{2,LR}$	$\hat{y}_{2,DT}$	$\hat{y}_{2,SVC}$	$y_{\text{Fold 2}}$
Fold 3	Folds 1, 2, 4, 5	**Fold 3 Rows**	$\hat{y}_{3,LR}$	$\hat{y}_{3,DT}$	$\hat{y}_{3,SVC}$	$y_{\text{Fold 3}}$
Fold 4	Folds 1, 2, 3, 5	**Fold 4 Rows**	$\hat{y}_{4,LR}$	$\hat{y}_{4,DT}$	$\hat{y}_{4,SVC}$	$y_{\text{Fold 4}}$
Fold 5	Folds 1, 2, 3, 4	**Fold 5 Rows**	$\hat{y}_{5,LR}$	$\hat{y}_{5,DT}$	$\hat{y}_{5,SVC}$	$y_{\text{Fold 5}}$
Full Matrix	-	**Concatenated OOF**	**Unbiased** Z_1	**Unbiased** Z_2	**Unbiased** Z_3	**Original y (Zero Leakage!)**

📊 Ma'am's 2-Level Stacking Architecture



🔢 Mathematical Formulation

The Level-0 Out-of-Fold (OOF) meta-feature matrix $\mathbf{Z} \in \mathbb{R}^{N \times M}$:

$$Z_{i,m} = \hat{f}_m^{(-k(i))}(x_i)$$

where $\hat{f}_m^{(-k(i))}$ is model m trained excluding the fold $k(i)$ containing observation i.

The Level-1 Meta-Learner optimization:

$$\min_{\mathbf{w}} \sum_{i=1}^{N} \mathcal{L}\left(y_i, g(\mathbf{z}_i; \mathbf{w})\right)$$

💻 Implementation

The fully tested and verified standalone implementation is provided in the synchronized Jupyter Notebook.